

Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
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Assignment 3(Partial Differential Equations)

1. Form the partial differential equation by eliminating the arbitrary constants from the following

i) $(x-a)^2 + (y-b)^2 + z^2 = c^2$ ii) $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$ iii) $z = a \log \left[\frac{b(y-1)}{1-x} \right]$

2. Form the partial differential equation by eliminating the arbitrary functions from the following

i) $f(x+y+z, x^2+y^2+z^2) = 0$ ii) $z = f(x+at) + g(x-at)$ iii) $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$

iv) $u = f(x^2+y) + g(x^2-y)$ v) $u = f(x+y)g(x-y)$ vi) $z = \frac{1}{x}f(y-x) + f'(y-x)$

3. If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that $(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$.

4. Solve the following PDEs by Lagrange's method

i) $(x+y^2)p + yq = z + x^2$ ii) $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$ iii) $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

iv) $\frac{y-z}{yz}p + \frac{z-x}{zx}q = \frac{x-y}{xy}$ v) $px(z-2y^2) = (z-xy)(z-y^2-2x^3)$ vi) $(z^2-2yz-y^2)p + (xy+zx)q = xy-zx$

5. Solve the following PDEs by standard forms

i) $q^2 = z^2 p^2 (1 - p^2)$ ii) $z^2 (p^2 x^2 + q^2) = 1$ iii) $z^2 (p^2 + q^2) = x^2 + y^2$ iv) $p^2 q^2 (px + qy - z) = 1$

6. Solve the following PDEs by Charpit's method

i) $2z + p^2 + qy + 2y^2 = 0$ ii) $pxy + pq + qy = yz$ iii) $(p^2 + q^2)y = qz$ iv) $2xz - px^2 - 2xyq + pq = 0$

7. Using the transformations $z^2 = u$, reduce the PDE $z^2(2p^2 + 3q^2) = 8x^2 + 27y^2$ to separable form and hence solve the equation.

8. Using the transformations $x^2 = u$, $y^2 = v$ reduce the PDE $2xyz = px^2y + qxy^2 + 4pq$ to Clairaut's form and hence solve the equation.

9. Find the general solution of the following PDEs

i) $2\frac{\partial^2 z}{\partial x^2} + 5\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 0$ ii) $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$

iii) $3r + 7s + 2t + 7p + 4q + 2z = 0$ iv) $(D^2 - 4DD' + 4D'^2)z = e^{2x+y}$

v) $(D^2 - DD' - 2D)z = \cos(3x + 4y)$ vi) $(D^2 + DD' - 6D'^2)z = \cos(2x + y)$

vii) $(D^2 + 2DD' + D'^2)z = x^2 + xy + y^2$ viii) $(D^3 + D^2D' - DD'^2 - D'^3)z = e^x \cos y$

ix) $(4D^2 - 4DD' + D'^2)z = 16 \log(x + 2y)$ x) $(D^2 - DD')z = \cos 2y(\sin x + \cos x)$

xi) $(D - D' - 1)(D + D' - 2)z = e^{2x-y}$ xii) $(6D^2 + 5DD' - 6D'^2)z = 132 \log(x + 3y)$

xiii) $(D + D' - 1)(D + 2D' - 3)z = 4 + 3x + 6y$ xiv) $(D^2 - D'^2 - 3D + 3D')z = xy + e^{x+2y}$

xv) $(D^2 - DD' - 2D'^2 + 2D' + 2D)z = e^{2x+3y} + \sin(2x + y) + xy$

xvi) $D(D + D' - 1)(D + 3D' - 2)z = x^2 - 4xy + 2y^2$

10. Classify the following PDEs

i) $u_{xx} = 5u_x + u_y$ ii) $u_{xx} + u_{yy} + 3u_{xy} = 0$ iii) $xu_{xx} + yu_{yy} = 0$

iv) $x^2u_{xx} + (1-y^2)u_{yy} = 0$ for $-1 < y < 1, -\infty < x < \infty$ vi) $u_{xx} + 4u_{xy} + (x^2 + 4y^2)u_{yy} = \sin(x+y)$

11. Using the method of separation of the variables, solve the the following PDEs

i) $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u, \quad u(x, 0) = 6e^{-3x}$ ii) $4\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 3u, \quad u(x, 0) = 3e^{-x} - 3e^{-5x}$

iii) $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = 4x - \frac{x^2}{2}$ iv) $\frac{\partial^2 u}{\partial t^2} = 16\frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = x^2(5-x)$

12. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin \left(\frac{\pi x}{l} \right) \cos \left(\frac{\pi ct}{l} \right)$.

13. Solve the one dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with the initial boundary conditions $u(x, 0) = 3 \sin(n\pi x), \quad u(0, t) = 0$ and $u(1, t) = 0$, where $0 < x < 1, \quad t > 0$.

14. Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the conditions $u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, a) = \sin \left(\frac{n\pi x}{l} \right)$.